# Giulia Rocco

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## About

**Biomedical Engineer and MSCA Fellow** specializing in multimodal neuroimaging (fNIRS, fMRI, EEG) and advanced signal processing. Expertise in brain functional mapping and the development of tailored analytical pipelines for high-dimensional neural data. Proven track record in leading international collaborations (Canada, France, Italy) and securing competitive funding. Current research focuses on calibrated fMRI and optical imaging to quantify cerebrovascular function and oxygen metabolism at rest and during cognitive tasks in healthy and clinical populations.

## Experience

**Institute for Advanced Biomedical Technologies**, Postdoctoral researcher — Chieti, Italy — Mar 2025 – present

- Multimodal data collection and analysis (fMRI, fNIRS) with healthy controls and multiple sclerosis patients
- Coordinated cross-institutional workflows, streamlining scientific and administrative milestones to ensure project deadlines were met

**I3S Lab and Centre Hospitalier Universitaire de Nice**, Graduate Research Assistant — Nice, France — Jan 2020 – Nov 2023

- Conducted multimodal data collection and analysis (fMRI, fNIRS, EEG) to investigate cerebellar activation, resulting in 6 published findings
- Designed and validated non-invasive novel experimental frameworks for studying cerebellum, earning awards for breakthroughs in cerebellar neuroscience
- Led cross-institutional project management, coordinating 50+ subject recruitment, 20k$ budget oversight, and data integrity, while also successfully obtaining two equipment loans from industry to facilitate experiments

**NIRx Medical Technologies**, PhD Intern — Berlin, Germany — June 2022 – July 2022

- Developed a customer-facing document detailing the value proposition of simultaneous fNIRS/fMRI, including hardware and software summaries, and drafted a video plan for data analysis software promotion

**Concordia University, MultifunkIM Lab**, Visiting Researcher — Montreal, Canada — Mar 2022 – Dec 2022

- Pioneered a computational framework for cerebellar fNIRS image reconstruction, effectively solving forward and inverse problems
- Led multimodal data collection projects involving 30 participants and 300+ recordings

**Université Côte d'Azur**, Teaching Assistant — Nice, France — Sept 2020 – Dec 2020

- Taught Algorithms & Programming with R to 30+ students, delivering practical sessions and project-based assignments

**Politecnico di Milano, B3Lab and PHEEL Laboratories**, Research Intern — Milan, Italy — Jan 2019 – Nov 2019

- Analyzed biomedical signals (EEG, ECG, PPG, EDA, eye tracking) for emotion and stress detection, developing novel indices for biomedical and biomarketing applications
- Contributed to multidisciplinary VR therapy research for neurodevelopmental disorders, managing physiological monitoring and data analysis
- Conducted a comparative analysis of two popular wearable devices and a clinical-grade ECG, revealing significant variations in inter-beat-interval accuracy

**KU Leuven, Audio Engineering Lab**, Visiting Scholar — Leuven, Belgium — Oct 2017 – Mar 2018

- Design and implementation of a novel language-independent speech intelligibility test for hearing screening, resulting in 3 publications
- Characterized speech stimuli intelligibility and psychometric functions with different noise realizations

**Lutech**, Support Engineer — Milan, Italy — Oct 2015 – Dec 2015

- Support to staff in operating rooms within the development of an EHR framework

**Università di Milano, LAFAS Lab**, Visiting Student — Milan, Italy — Mar 2015 – July 2015

- Validated and optimized machine learning algorithms for CT image segmentation, achieving a correlation of over 0.9 with manual methods and reducing processing time by 53%

## Education

**Université Côte d'Azur**, *Nice, France* — PhD in Automatics, Signal and Image processing — Jan 2020 – Nov 2023
- Marie Curie fellow - **BoostUrCAreer** program
- Thesis: *"Multimodal Functional Exploration of the Cerebellum"*

**KTH Royal Institute of Technology**, *Stockholm, Sweden* — MSc in Biomedical Engineering — Aug 2016 – Mar 2017
- Erasmus+ Exchange

**Politecnico di Milano**, *Milan, Italy* — MSc in Biomedical Engineering — Sept 2015 – Apr 2018
- Final Grade: 110/110
- Thesis Research at KU Leuven: *"Design, implementation, and pilot testing of a language-independent speech intelligibility test"*

**Politecnico di Milano**, *Milan, Italy* — BSc in Biomedical Engineering — Sept 2012 – July 2015
- Final Grade: 102/110
- Thesis: *"Validation and optimisation of an automatic hard tissue segmentation algorithm for CBCT data"*

## Skills

**Programming:** Matlab, R, C++, Python, Bash, Git

**Data analysis:** SPM, FreeSurfer, FSL, AFNI, Brainstorm, Fieldtrip, Homer3, EEGLAB

**Languages:** Italian (native), English, French

## Funding

**2026**: Marie Curie General Microgrant (1000€)

**2022**: Quebec Bio-Imaging Network grant for Pilot Projects ($20,000) - Conceived the project and drafted the application

**2020**: Horizon 2020 MSCA COFUND doctoral scholarship (42 months) - 15 students selected over 500 applicants

**2019**: Biomedical Engineering departmental funds for research interns, Politecnico di Milano
Merit scholarship for master thesis abroad, Politecnico di Milano

**2017**: Erasmus+ mobility scholarship (2 spots for KTH for Biomedical Engineering), Politecnico di Milano

**2015**: Merit scholarship for off-campus students, Politecnico di Milano

## Awards

**2026**: Trainee Abstract Award, Brain Function Study Group, International Society for Magnetic Resonance in Medicine

**2023**: Nomination for the best PhD thesis prize of the doctoral school

**2021**: Public Award, 3-minutes Thesis contest in 18th IEEE International Symposium on Biomedical Imaging
Best Poster Award, 4th Congress of the Physiology and Integrative Biology French Society
Winner, MSCA Falling Walls Lab 2021: Breaking the wall of Cerebellar Neuroscience (Top 75 Emerging Talents worldwide)
Université Côte D'Azur Excellence Prize

**2012**: Winner, Literary Contest - An Author to Rediscover – Sibilla Aleramo: A Journey Called Woman
Winner, Latin Contest - Certamen Leonianum
Honorable Mention, Latin Contest - Certamen Horatianum

## Teaching

**2025**: Teaching Assistant, Industrial Bioengineering & Bioengineering and Biomenchanics, University of Chieti-Pescara
Teaching Assistant, Physics, University of Chieti-Pescara

**2024**: Co-instructor, "NIRSTORM: a Brainstorm plugin for fNIRS statistical analysis", Educational course at SfNIRS 2024

**2022**: Co-instructor, "NIRSTORM: a Brainstorm plugin for fNIRS statistical analysis", Educational course at SfNIRS 2022

**2020**: Teaching Assistant, Algorithms & Programming with R, Université Côte D'Azur

## Talks and Presentations

**2026**: **Invited lecture**, Methods for Multimodal Functional Evaluation of the CNS, Politecnico di Milano

*A perspective on simultaneous fNIRS/fMRI brain measurements*

**Poster**, International Society for Magnetic Resonance in Medicine (ISMRM)

*Simultaneous gas-free calibrated fMRI and fNIRS measurements in human visual cortex: a feasibility study*

**2025**: **Poster**, fNIRS UK Biennial meeting

*Simultaneous fNIRS, BOLD, and CBF Measurements on the Visual Cortex: a feasibility study*

**Poster**, fNIRS Italy Biennial meeting

*Toward simultaneous calibrated fMRI and fNIRS acquisitions: a case study in the visual cortex*

**2024**: **Oral talk**, Biennial meeting of the Society of fNIRS (SfNIRS)

*Enhancing Cerebellar fNIRS/fMRI via Tailored Pipelines*

**Invited talk**, RWTH-NIRx fNIRS workshop

*Simultaneous fNIRS/fMRI exploration of the cerebellum*

**2022**: **Poster**, Biennial meeting of the Society of fNIRS (SfNIRS)

*Densifying Optodes Montage to Enhance Cerebellar fNIRS*

**Poster**, 19th IEEE International Symposium on Biomedical Imaging (ISBI)

*When fNIRS meets fMRI to complement cerebellar exploration*

**2021**: **Poster**, 4th Congress of the Physiology and Integrative Biology French Society

*Asymmetries in Cerebellar Activation during Finger Movements: A Functional Near-Infrared Spectroscopy Study*

**Paper presentation**, 43rd IEEE Engineering in Medicine and Biology Society (EMBC)

*A chiral fNIRS spotlight on cerebellar activation in a finger tapping task*

**Poster**, 18th IEEE International Symposium on Biomedical Imaging (ISBI)

*Exploring the cerebellum with functional near-infrared imaging: a preliminary study*

**2020**: **Paper presentation**, 42nd IEEE EMBC

*Exploration of the physiological response to an online gambling task by frequency domain analysis of the electrodermal activity*

**Poster**, SophIA Summit

*P300 event-related potentials classification from EEG data through interval feature extraction and recurrent neural networks*

## Leadership and Service

### REVIEWER

- IEEE Transactions on Neural Systems and Rehabilitation Engineering
- International Journal of Audiology

### SUPERVISION

- MSc: 1. BSc: 1

### ORGANIZATION

- fNIRS Italy Biennial meeting 2027, Pescara (Organising Committee)
- BoostUrCareer Doctoriales 2021, online (Program Committee)

### SERVICE

- Biennial meeting of the Society of Functional Near-Infrared Spectroscopy (SfNIRS) 2026, Macau (Conference support team)
- MSCA Falling Walls Lab 2022 (Jury member)

## Professional Formation

**2023**: Introduction to ethics and scientific integrity

**2022**: PhD Career development, Association Bernard Gregory (ABG)

**2021**: TED-worthy presenting, Debatrix - funded by MSCA Falling Walls Lab

BootCamp Open Science, OPTIMA, TU Graz - Best practices

## Outreach

**2025**: European Researcher Night, University of Chieti-Pescara - The physics behind MRI
Tales from MEDNIGHT - Translator of short science-inspired tales for kids

**2024**: Tales from MEDNIGHT - Translator of short science-inspired tales for kids

**2023**: Brain Week, Université Côte D'Azur - Introduction to brain measurements
Secondary School Outreach Mentor, Université Côte D'Azur - Practical laboratory on optical imaging

**2022**: International Day of Women and Girls in Science - Interviewed by MSCA, European Commission
Secondary School Outreach, Université Côte D'Azur - Introduction to brain function and optical imaging

**2021**: European Researcher Night, Université Côte D'Azur - Researcher's suitcase

**2020**: European Researcher Night, Université Côte D'Azur - Science comics

## Publications

**A comparison of continuous-wave fNIRS with quantitative fMRI-derived indices of brain function in the visual cortex** — *In preparation*
Rocco, G., Chalet, L., Fear, E. J., Pomante, S., Graziano, F., Di Censo, D., Carriero, M., Delaire, E., Esposito, F., Perrucci, M. G., Del Gratta, C., Perpetuini, D., Wise, R., Chiarelli, A. M.

**Simultaneous fNIRS-fMRI in Cerebellum and Motor Cortex Reveals High Temporal Correlation with Pipeline-Dependent Spatial Variability** — *In preparation*
Rocco, G., Delaire, E., Ramanoël, S., Meste, O., Magnié-Mauro, M. N., Grova, C., Lebrun, J.

**Surface-based Analysis of Cerebellar Personalized High-Density fNIRS and fMRI During a Motor and an Auditory Task** — *In preparation*
Rocco, G., Delaire, E., Cai, Z., Ramanoël, S., Meste, O., Magnié-Mauro, M. N., Grova, C., Lebrun, J.

**Three-dimensional T1 mapping demonstrates the transfer of oxygen into cerebrospinal fluid during hyperoxia** — *Under review*
Biondetti, E., Di Censo, D., Pomante, S., Censi, S., Bliakharskaia, E., Carriero, M., Chalet, L., Rocco, G., Buonincontri, G., Graziano, F., Caporale, A., Chiarelli, A. M., Wise, R.

**Quantitative functional BOLD (qfBOLD): A Gradient-Echo/Spin-Echo Framework for Oxygen Extraction Fraction (OEF) Mapping with functional MRI** — May 2026
Chiarelli, A. M., Chalet, L., Pomante, S., Di Censo, D., Caporale, A., Biondetti, E., Fasano, F., Zaca', D., Rocco, G., Carriero, M., Graziano, F., Fear, E. J., Caligiuri, M. E., Wise, R., Germuska, M.
(Journal of Cerebral Blood Flow and Metabolism)

**Simultaneous gas-free calibrated fMRI and fNIRS measurements in human visual cortex: a feasibility study** — May 2026
Rocco, G., Chalet, L., Pomante, S., Fear, E. J., Graziano, F., Di Censo, D., Carriero, M., Perpetuini, D., Del Gratta, C., Perrucci, M. G., Wise, R., Chiarelli, A. M.
(International Society for Magnetic Resonance in Medicine (ISMRM))

**Assessing the Minimum Number of Breath-Holds Required for Stable $CMRO_2$ and OEF Estimates Using Breath-Hold Calibrated fMRI** — May 2026
Fear, E. J., Di Censo, D., Pomante, S., Carriero, M., Graziano, F., Chalet, L., Rocco, G., Caporale, A., Biondetti, E., Fasano, F., Zaca', D., Tomassini, V., Chiarelli, A. M., Wise, R.
(International Society for Magnetic Resonance in Medicine (ISMRM))

**Mapping Cerebrovascular Reactivity: Comparison of Gradient Echo and Spin Echo BOLD with Arterial Spin Labelling** — May 2026
Pomante, S., Di Censo, D., Caporale, A., Fasano, F., Zaca', D., Fear, E. J., Graziano, F., Carriero, M., Rocco, G., Chalet, L., Biondetti, E., Caligiuri, M. E., Germuska, M., Wise, R., Chiarelli, A. M.
(International Society for Magnetic Resonance in Medicine (ISMRM))

**Assessing Cerebral Oxygen Metabolism in Multiple Sclerosis Using Breath-Hold single Calibrated fMRI** May 2026
Di Censo, D., Fear, E. J., Caporale, A., Censi, S., Graziano, F., Biondetti, E., Chalet, L., Rocco, G., Carriero, M., Pomante, S., Patitucci, E., Fasano, F., Zaca', D., Germuska, M., Chiarelli, A. M., Tomassini, V., Wise, R.
(International Society for Magnetic Resonance in Medicine (ISMRM))

**Quantitative Functional BOLD (qfBOLD) for Oxygen Extraction Fraction Mapping Using ASL-free Gradient Echo/Spin Echo fMRI** May 2026
Chiarelli, A. M., Chalet, L., Pomante, S., Di Censo, D., Caporale, A., Biondetti, E., Fasano, F., Zaca', D., Rocco, G., Carriero, M., Graziano, F., Fear, E. J., Caligiuri, M. E., Wise, R., Germuska, M.
(International Society for Magnetic Resonance in Medicine (ISMRM))

**Integrating a flow-diffusion model of O2 transport to qBOLD for contrast-agent- and gas-free mapping of deoxyCBV and CMRO2** May 2026
Chalet, L., Di Censo, D., Pomante, S., Rocco, G., Fear, E. J., Graziano, F., Carriero, M., Fasano, F., Biondetti, E., Perpetuini, D., Wise, R., Chiarelli, A. M.
(International Society for Magnetic Resonance in Medicine (ISMRM))

**Long Short-Term Memory Networks for Fast Optical Signal Identification in the Human Visual Cortex for Brain Computer Interface Applications** Nov 2025
Perpetuini, D., Rocco, G., Fear, E., Pomante, S., Chalet, L., Graziano, F., Valeri, E., Carriero, M., Del Gratta, C., Perrucci, M. G., Wise, R., Chiarelli, A. M.
(International Conference on e-Health and Bioengineering (EHB))

**Enhancing Cerebellar fNIRS/fMRI via Tailored Pipelines** Sept 2024
Rocco, G., Delaire, E., Ramanoël, S., Meste, O., Magnié-Mauro, M. N., Grova, C., Lebrun, J.
(Biennial meeting of the Society of Functional Near-Infrared Spectroscopy (SfNIRS))

**Characterization of the Intelligibility of Vowel–Consonant–Vowel (VCV) Recordings in Five Languages for Application in Speech-in-Noise Screening in Multilingual Settings** May 2023
Rocco, G., Bernardi, G., Ali, R., van Waterschoot, T., Polo, E. M., Barbieri, R., Paglialonga, A.
(Applied Sciences)

**Densifying Optodes Montage to Enhance Cerebellar fNIRS** Oct 2022
Rocco, G., Delaire, E., Ramanoël, S., Meste, O., Magnié-Mauro, M. N., Grova, C., Lebrun, J.
(Biennial meeting of the Society of Functional Near-Infrared Spectroscopy (SfNIRS))

**When fNIRS meets fMRI to complement cerebellar exploration** Mar 2022
Rocco, G., Ramanoël, S., Habas, C. N., Arleo, A., Meste, O., Magnié-Mauro, M. N., Lebrun, J.
(IEEE International Society for Biomedical Imaging (ISBI))

**A chiral fNIRS spotlight on cerebellar activation in a finger tapping task** Nov 2021
Rocco, G., Lebrun, J., Meste, O., Magnié-Mauro, M. N.
(43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC))

**Single-Trial Detection of Event-Related Potentials with Integral Shape Averaging: An Application to the Elusive N400** Nov 2021
Rocco, G., Rix, H., Lebrun, J., Guetat, S., Chanquoy, L., Meste, O., Magnié-Mauro, M. N.
(43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC))

**Asymmetries in Cerebellar Activation during Finger Movements: A Functional Near-Infrared Spectroscopy Study** Sept 2021
Rocco, G., Lebrun, J., Meste, O., Magnié-Mauro, M. N.
(Acta Physiologica: 4th Congress of Physiology and Integrative Biology)

**Exploring the cerebellum with functional near-infrared imaging: a preliminary study** Apr 2021
Rocco, G., Lebrun, J., Meste, O., Magnié-Mauro, M. N.
(IEEE 18th International Society for Biomedical Imaging (ISBI))

**P300 event-related potentials classification from EEG data through interval feature extraction and recurrent neural networks** Nov 2020
Rocco, G., Lebrun, J., Magnié-Mauro, M. N., Meste, O.
(SophIA Summit)

**An automated speech-in-noise test for remote testing: Development and preliminary evaluation** Sept 2020
Paglialonga, A., Polo, E. M., Zanet, M., Rocco, G., van Waterschoot, T., Barbieri, R.
(American Journal of Audiology)

**Exploration of the physiological response to an online gambling task by frequency domain analysis of the electrodermal activity** July 2020
Rocco, G., Reali, P., Lolatto, R., Tacchino, G., Mandolfo, M., Mazzola, A., Bianchi, A. M.
(42nd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC))

**Characterization of eye gaze and pupil diameter measurements from remote and mobile eye-tracking devices** Jan 2020
Lolatto, R., Rocco, G., Mustoni, R., Maninetti, C., Pastura, R., Pigazzini, A., Barbieri, R.
(XV Mediterranean Conference on Medical and Biological Engineering and Computing (MEDICON))

**Development and preliminary evaluation of a novel adaptive staircase procedure for automated speech-in-noise testing** July 2019
Zanet, M., Polo, E. M., Rocco, G., Paglialonga, A., Barbieri, R.
(41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC))

**Heart rate variability from wearables: a comparative analysis among standard ECG, a smart shirt and a wristband** June 2019
Reali, P., Tacchino, G., Rocco, G., Cerutti, S., Bianchi, A. M.
(pHealth 2019)

**Validation of an automatic hard tissue segmentation algorithm for cone beam CT data** Aug 2015
Codari, M., Caffini, M., Rizzo, L., Rocco, G., Tartaglia, G. M., Baselli, G., Sforza, C.
(37th IEEE Annual conference on Engineering in Medicine and Biology)